

CALCULUS 101 · RECITATION HOMEWORK

Homework 1: How Fast Is the Baby Growing?

Name: _____

Date: _____

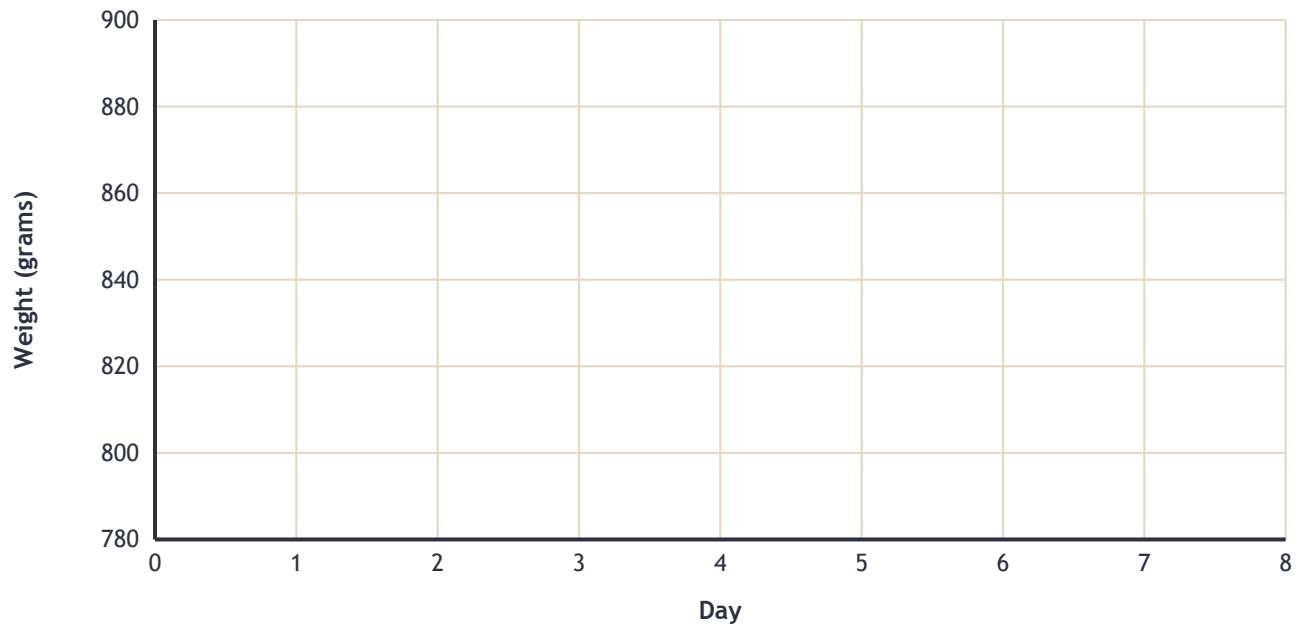
Instructions: About 20 minutes. Rate of change = change $\div$ time. Show your subtraction and division — the steps are worth points too. Total: **20 points**.

Part A — Rates from a table

1. Three babies were weighed on day 1 and again on day 8. Fill in the change in grams, the number of days, and the rate in grams per day. (Change = later weight - earlier weight. Rate = change $\div$ days.) (6 pts)

Baby	Day 1 (g)	Day 8 (g)	Change (g)	Days	Rate (g/day)
A	800	870			
B	1,200	1,340			
C	2,000	2,000			

Part B — Plot it



2. On the grid above, plot Baby A's weights as dots and connect them: day 1 — 800 g, day 3 — 820 g, day 5 — 840 g, day 7 — 860 g. Then answer: is the line steep or gentle? What is Baby A's rate in grams per day? (3 pts)

Part C — When the line goes down

3. A full-term newborn weighed 3,400 g at birth and 3,200 g on day 4. (a) What was the change? (b) What was the rate per day? (c) Should the team worry? (Hint: newborns normally lose up to about 10% of their birth weight in the first days, then regain it.) (3 pts)

4. On a weight graph, a steeper line means... (1 pt)

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- A. the weight is changing faster.
- B. the baby weighs more.
- C. more days have gone by.
- D. the baby is older.

Part D — Per kilogram

5. A baby weighs 2 kg and gains 40 g in one day. What is that in grams per kilogram per day (g/kg/day)? Is it in the healthy 15-20 range? (2 pts)

Part E — Flashback

6. From NICU Tutor: what is the healthy growth target for a preemie, in g/kg/day? (1 pt)

7. From Biology 101: one cell goes through 3 rounds of division. How many cells now? (1 pt)

Part F — Think like a doctor

8. Two preemies both weigh exactly 1,500 g today. Baby X's weight line has climbed steadily all week. Baby Y's line has been flat for a week. Which baby worries you more, and what might the team do about it? (3 pts)

Score: _____ / 20 points

Professor's comment:
